

Lecture 32

4/13/26
TRA

Wednesday, started last 2 techniques

Did side-by-side comparison of x-rays & nuclear

X-rays	- Transmission based	anatomical
Nuclear	- emission based	Functional

Many machines do both techniques simultaneously

RNI is a hot technique now
Personalized

Patient is given a "radiopharmaceutical"
A radioactive ^{cpd} (element) attached to a
pharmaceutical (e.g., a glucose analog)

Idea - pharm goes preferentially to spot(s)
of interest in body
Glucose → tumors due to higher metabolic
activity

Radioactive glucose accumulates in tumor
(trapped).

Radioactive nucleus "emits" ^{high-energy photons} γ-ray photons

The γ -rays can escape body & are used to create an image

Strengths - Functional

- high-contrast

Negatives - ionizing radiation

- poor resolution

A little about nuclear instability & radioactivity

Nucleus = protons + neutrons

Elements can exist in different isotopic forms

- # protons determines elements

- # neutrons can vary

Many isotopes are unstable, which is what we want to do radionuclide imaging

Not the best state energetically

Nucleus tries to change to become more stable $\Rightarrow$ undergoes radioactive decay

Unstable parent $\rightarrow$ more stable daughter

Why would a nucleus be unstable

Nucleus is really tiny w/ many protons crammed in it $\Rightarrow$ tons of repulsion

$\Rightarrow$ Lot of repulsion Nucleus should blow up

Need a counteracting attractive force to balance $\Rightarrow$ strong nuclear force that stabilizes many nuclei

Nevertheless, many nuclei are unstable

(Large nuclei always unstable)

Nuclear medicine purposefully uses unstable nuclei

Various decay routes

Electrons
Positrons
α
β
Positron

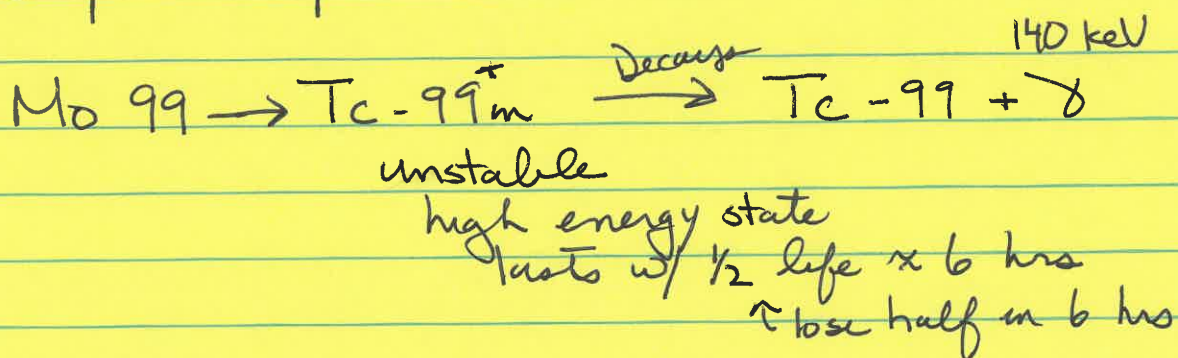
Goal is to get γ rays that can escape body

Dominant radionuclides used in nuclear medicine

~~Imp +~~
~~Biggest~~ (1) $Tc-99m$ ← metastable
↑
Technetium

Nuclear decay scheme for $Tc-99m$

A lot of mechanisms / schemes involve multiple steps



High energy state analogous to excited state in fluorescence

Call photon a γ -ray because it comes from nuclear decay

(2) Another prominent example for nuclear medicine is ^{18}F Fluorine

^{19}F Fluorine is the stable version

↙ positron emitter

^{18}F has a low neutron/proton (N/p) ratio

To fix this problem, nucleus must shed or emit a positive charge in the form of a positron



$\downarrow P$ $N/p \uparrow$
 $\uparrow N$
 $\Rightarrow$ improved stability

In contrast, if high N/p ratio, emit an electron

Half-life $T_{1/2} = 110 \text{ min} \approx 2 \text{ hrs}$

Half gone every 2 hrs

Where are the γ rays? Super impt in med

The techniques are similar to X-ray techniques Prog + Tomog

^{18}F decays $\hat{=}$ makes e^+

e^+ quickly finds an e^- in about $\sim 0.1 \text{ mm}$

$e^+ + e^-$ matter $\hat{=}$ anti-matter
annihilates

Particles disappear, & their energy is converted into γ -rays (EM radiation)

$$E = mc^2$$

e^+, e^-

if pos i. electron is
Energy equivalent
= 511 keV

Two γ rays emitted

$e^+ e^-$
at rest

Need two because $e^+ e^-$ approximately at rest, so momentum at start is zero, so momentum must be zero after.

$\Rightarrow$ need two anti-parallel γ rays



This is key to PET technique
Basis of PET

What are useful decay times ($\frac{1}{2}$ lives) $T_{1/2}$ in medicine?

$T_{1/2}$ = time for $\frac{1}{2}$ of radioactive nuclei to decay

Best is on the order of hours for nuclear medicine

Tc-99m

$\tau_{1/2} \approx 6 \text{ hrs}$

cat
example

^{18}F

110 min

Needs to be fast enough to give a good imaging signal. Also want patient to become non radio

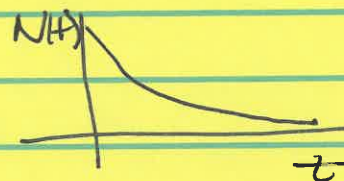
Fast enough so patient is NOT radioactive too long

Needs to be slow enough for radio pharm to get to site & for doctor to do scan before it decays significantly

radioactive nuclei decays exponentially

$$N(t) = N_0 e^{-\lambda t}$$

$\uparrow$ # of radioactive nuclei at time t
 $\uparrow$ starting #



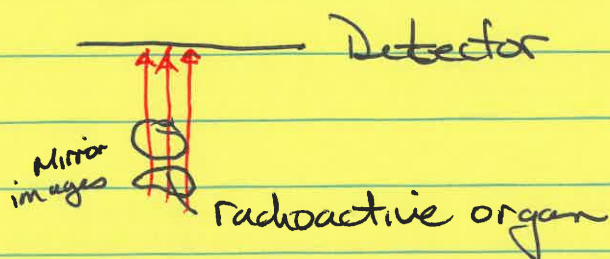
$$\tau_{1/2} = \frac{0.693}{\lambda}$$

Two modalities in RNI

Like
X-ray
based
imaging

Projection & Tomographic techniques

Projection $\leftarrow$ Lack depth info
 useful but Suffer from structural superposition



No difference in projection

Sometimes projections are good enough

Special problem w/ radionuclide projections
 $\rightarrow$ rays emitted isotropically

Big problem in RNI

Is are emitted isotropically

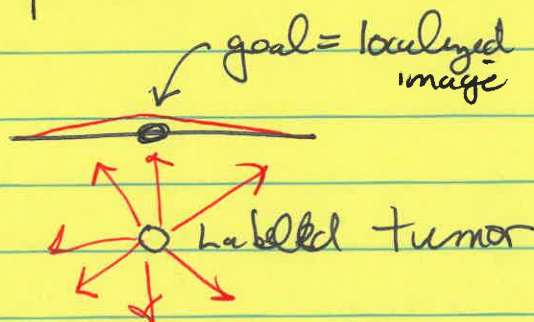


Image will be spread out over whole detector

Overcome this w/ a "collimator"

Collimator is used to solve this problem

Collimator lowers or reduces signal a lot but it creates a meaningful projection

correspondence between location of tumor & projection

so markedly reduces signal $\Rightarrow$ noisy images

PET does NOT use collimator due to two α -parallel gamma